

Benchmarking and Performance Analysis of LLM Inferencing Systems

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1. Various Scheduler Designs
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First Come First Serve Policy



- Adds requests one by one, from the core engine into a waiting queue
- The requests are extracted in a first come first serve manner
- We stop selecting requests from the waiting queue into the running batch, once the size reaches `max_batch_size`

Priority Based



- The user passes priority values along with the requests' strings themselves
- We use a min heap to find the request with the smallest priority value (executed at first)
- Ties are broken in a first come first serve manner (through tracking request's arrival time)

Greedy First Come First Serve



- We keep a track of the working set along with the requests we add to the current running scheduler batch
- We add requests in a first come first serve manner
- However, now we are also controlling the batch size - we stop whenever addition of the next request increases the working set size beyond a threshold (decided by L2 cache size)
- For our workload, and GPU - it boils down to stopping whenever the next request belongs to a different family

Greedy First Come First Serve - Appendix - Expected Batch Size



Let's say we have N families. The user passes a bunch of requests, which can belong to any of these families. Moreover, all the requests have been shuffled, such that the probability of a request at position i to be of family j is exactly $\frac{1}{N}$.
Assume that the first request belongs to family j. We find out the expected value of the size of the first scheduler batch.
For the batch to have size exactly x:

- The first $x-1$ requests after the initial one must also be of family j.
- The x-th request must be of a different family.

The probability that the batch is of size x is given by:

$$\Pr[X = x] = \left(\frac{1}{N}\right)^{x-1} \cdot \frac{N-1}{N}, \quad x \geq 1.$$

Greedy First Come First Serve - Appendix - Expected Batch Size



Let X denote the size of the first batch. Its expected value is:

$$\mathbb{E}[X] = \sum_{x=1}^{\infty} x \Pr[X = x] = \sum_{x=1}^{\infty} x \left(\frac{1}{N}\right)^{x-1} \frac{N-1}{N}.$$

Using the identity

$$\sum_{x=1}^{\infty} x r^{x-1} = \frac{1}{(1-r)^2}, \quad |r| < 1,$$

with $r = \frac{1}{N}$, we obtain:

$$\mathbb{E}[X] = \frac{N-1}{N} \cdot \frac{1}{\left(1 - \frac{1}{N}\right)^2} = \frac{N}{N-1}.$$

$$\boxed{\mathbb{E}[\text{batch size}] = \frac{N}{N-1}}$$

For example, when we have 2 families, the expected batch size would be just 2.

Family ID Based



- If we can set priority values for each of the request, then why can't we set the family to which they belong to?
- We have separate queues for each family
- All the individual queues operate in a first come first serve manner
- If there is a request of family ID j already present in the scheduler batch, then we try to add all the requests of that family in the same batch (till we hit `max_batch_size`). If there is no request of that family available, then we stick to the current batch (no more requests added)
- So, the batch size is no longer static - it is dynamically set in accordance with the workload

Radix Tree Based



- Key idea: Instead of treating each request independently, we exploit shared token prefixes across requests
- Each request is represented as a path of tokens in a radix tree
- Internal nodes correspond to shared token substrings
- The scheduler maintains a working set of active nodes (tokens already materialized / cached)
- Each waiting request has a cost = number of tokens missing from the working set

Radix Tree Based



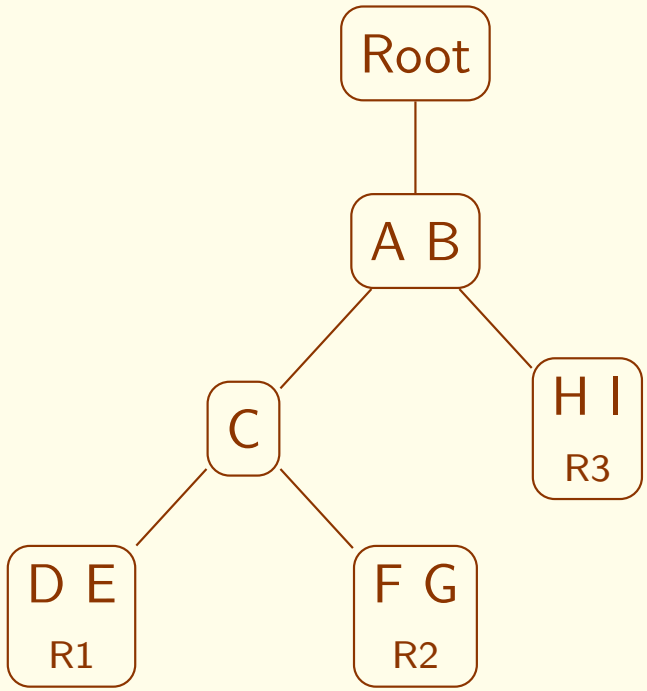
To schedule the next request:

- We start from leaves already in the working set
- Traverse upwards toward the root
- At each node, we look for waiting requests passing through it
- We select the request with minimum missing-token cost

Activating a request:

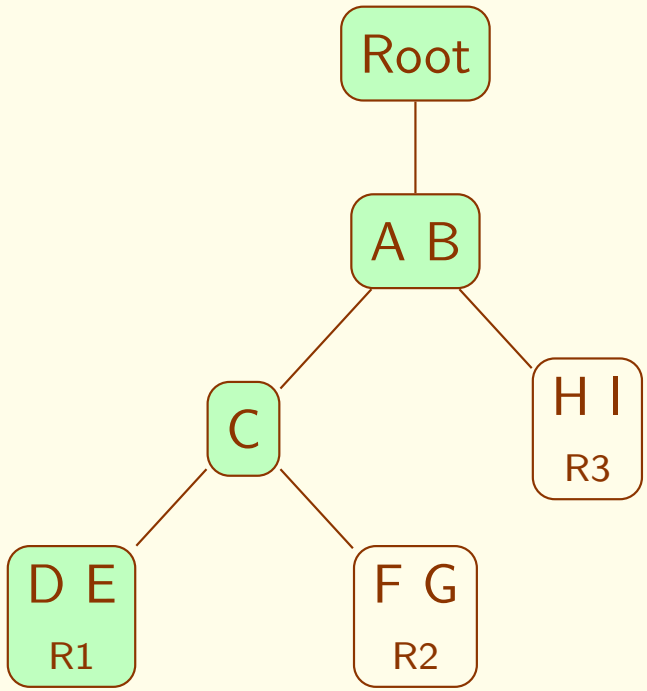
- Adds its path to the working set
- Future requests that share prefixes become cheaper
- This dynamically forms prefix-coherent batches, minimizing recomputation

Radix Tree Based - Radix Tree Construction



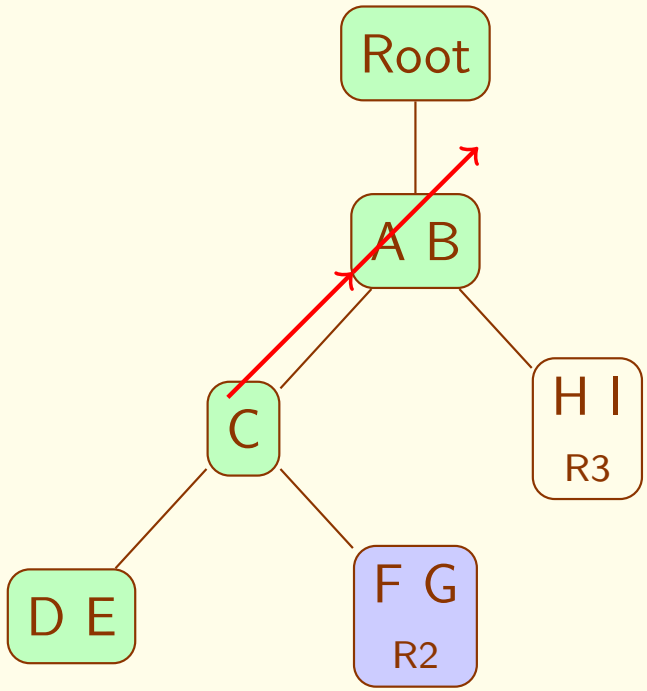
Each leaf corresponds to a request. Internal nodes represent shared token prefixes.

Radix Tree Based - Working Set After Activation



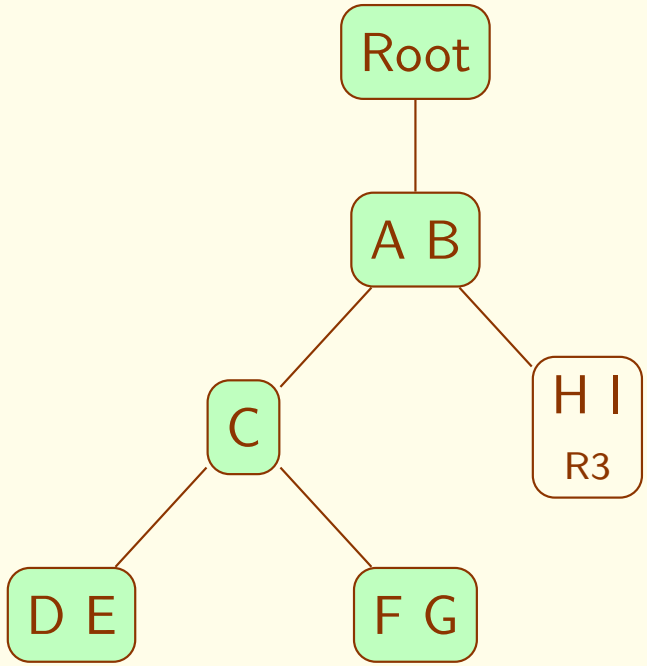
Green nodes indicate the working set after activating R1.

Radix Tree Based - Finding the Best Request



Traversal starts from working-set leaves and moves upward. R2 is selected due to maximum prefix overlap.

Radix Tree Based - Dynamic Batch Growth



Batch size grows dynamically as long as marginal cost is low.

Radix Tree Based - Issues



The above approach has a few issues:

- Adding a request is quite costly - there might be splitting/merging of nodes. Although finding the best request is quite efficient
- Each request may contain thousands of tokens, so the tree depth might get quite large too
- We cannot have a parallel thread/process handling the radix tree, because maintaining sync is difficult. What if we removed a request, from our waiting queue, but it takes some time to remove it from the radix tree?

Chunked Hash Tree Based



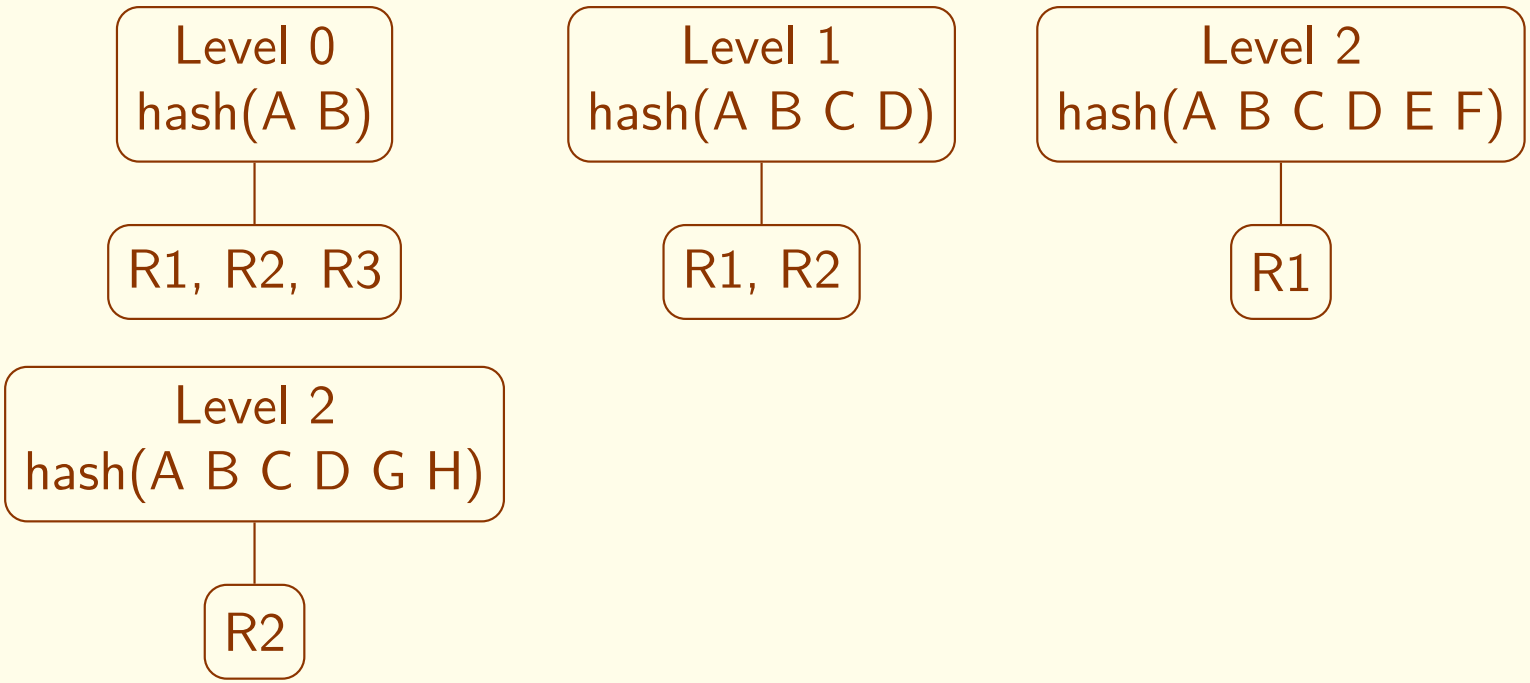
- Requests are represented using **hashed cumulative prefixes**
- Tokens are divided into fixed-size chunks
- Each level corresponds to `hash(tokens[0:k])`
- No explicit tree structure or node splitting
- Much faster than radix trees for large workloads

Chunked Hash Tree Based - Working Set



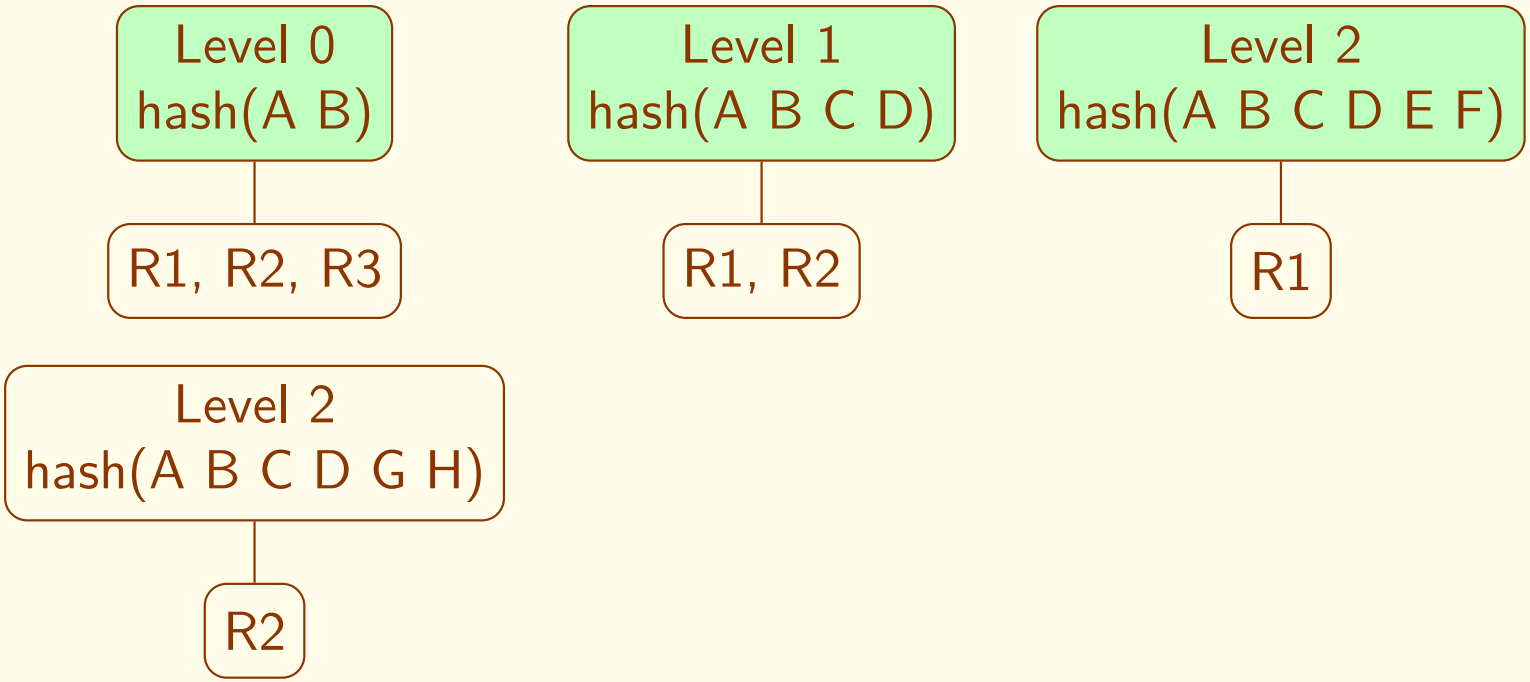
- Working set = set of active **(level, hash)** pairs
- Each active request contributes all its prefix hashes
- Request cost = number of missing prefix hashes
- Scheduling prefers requests with minimal missing chunks
- Enables dynamic batch growth similar to radix trees

Chunked Hash Tree Based - Insertion



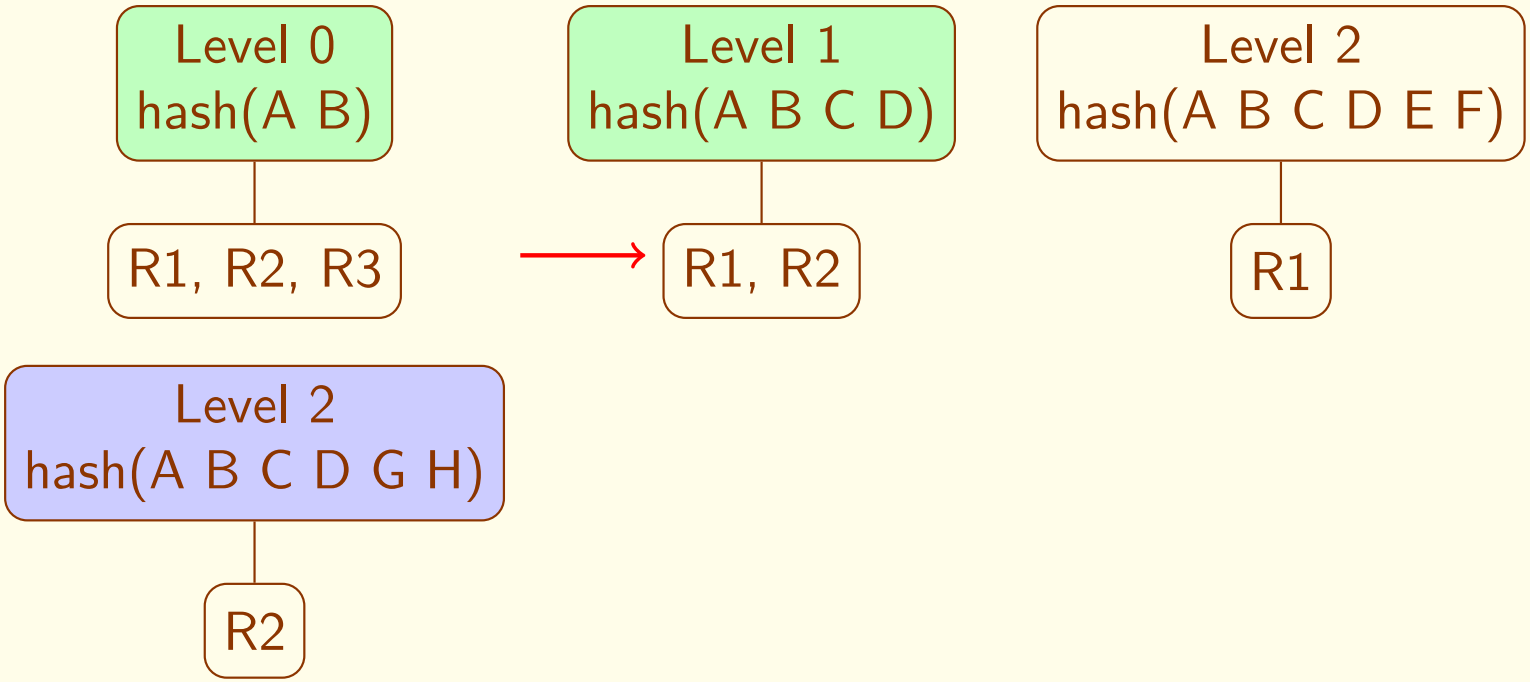
Each level stores hashes of cumulative token prefixes.

Chunked Hash Tree Based - Working Set



Green boxes indicate prefix hashes in the working set.

Chunked Hash Tree Based - Best Request Selection



R2 is selected since it has maximum prefix overlap with the working set.

Dynamic Batching



- Static batching uses a fixed maximum batch size
- Dynamic batching adapts batch size based on **runtime GPU memory state**
- Memory usage is dominated by KV cache allocation
- Larger batches improve throughput but increase memory pressure

Dynamic Batching - Memory-Aware Control



- Let B be the current batch size
- Measure used GPU blocks and estimate blocks per sequence:

$$\text{blocks_per_seq} = \frac{\text{used GPU blocks}}{B}$$

- Compute the memory-safe maximum batch size:

$$B_{\max} = \left\lfloor \frac{\text{total GPU blocks} \times \alpha}{\text{blocks_per_seq}} \right\rfloor$$

- α is a safety factor (e.g., $\alpha = 0.95$)
- Stop adding requests when $B \geq B_{\max}$

BucketServe - Bucket-Based Dynamic Batching



- Real-world workloads exhibit **highly heterogeneous sequence lengths**
- Naive batching mixes short and long requests, causing:
  - Excessive padding
  - GPU memory waste
  - Lower effective throughput
- Key idea: **group requests by sequence length** before batching
- Enables tighter memory bounds and more predictable batch sizing

BucketServe: Design



- Incoming requests are assigned to **length-homogeneous buckets**
- Each bucket covers a sequence length interval:

$$[L_b, U_b)$$

- Buckets reduce padding by minimizing (S is the sequence length):

$$S_{\max} - S_{\text{avg}}$$

- Dynamic batching is applied **within each bucket**
- Batch size is adjusted based on:
  - GPU memory availability
  - Maximum sequence length in the bucket

BucketServe : Memory Safety



- Bucket boundaries are adjusted online based on workload distribution
- Buckets are:
  - **Split** when length variance is high
  - **Merged** under low load to reduce scheduling overhead
- For each bucket, compute a memory-safe batch size:

$$N_{\max} = \max \left\{ N \;\middle|\; \sum_{i=1}^{N} S_i \leq M_{\text{safe}} \right\}$$

- Not too relevant for our workload, because the query sizes are roughly similar, and system prompt is anyway shared, so the expected throughput is quite similar to that of vLLM baseline

Workload Setup



- Two families (each of system prompt of length 1300 tokens)
- 2500 queries of each family - total 5000 queries
- All randomly shuffled
- L2 Cache size = 96 MB (RTX 6000 Ada Generation)
- Each family's system prompt's KVCache roughly takes 180 MB (model: Llama3.1-8B)
- We measure end-to-end throughput (including the time taken to add/process requests)
- We generate 50 new tokens for each request

Why just 50 new tokens?



- LLMs hardly answer in 1-2 sentences, so less than 50 (like somewhere around 10-20) does not make much sense
- The scheduler changes help increasing L2 purity, even during decodes, so larger number of output tokens show better throughput improvement
- Having 200 new tokens for a shared prefix of length 1300 tokens reduces the amount of overlap between two queries

Throughput



Scheduler	Throughput (req/sec)	Comments
First Come First Serve	45.49	vLLM Baseline
First Come First Serve - Family Ordered	64.97	User orders the requests by their families ; theoretical maximum which a user can get from vLLM through client side changes
Greedy First Come First Serve	2.04	Due to very small batch size
Family ID Based	86.72	The fastest which we can achieve ; although through demanding additional information from the user
Radix Tree Based	53.06	Inefficient design ; too much CPU overhead ; consumes a lot of time in between scheduler iterations
Chunked Hash Tree Based	77.44	A much more efficient design ; quite less overhead ; no extra information needed from the user
Dynamic Batching	46.98	Max batch size set to 1000

Table: Throughput Across Different Scheduler Designs

Throughput (for max new tokens = 100)



Scheduler	Throughput (req/sec)
First Come First Serve	20.84
First Come First Serve - Family Ordered	33.94
Family ID Based	42.73
Chunked Hash Tree Based	43.20

Table: Throughput Across Different Scheduler Designs

Throughput (for max new tokens = 200)



Scheduler	Throughput (req/sec)
First Come First Serve	10.67
First Come First Serve - Family Ordered	18.78
Family ID Based	21.14
Chunked Hash Tree Based	21.34

Table: Throughput Across Different Scheduler Designs

Throughput vs sequence length

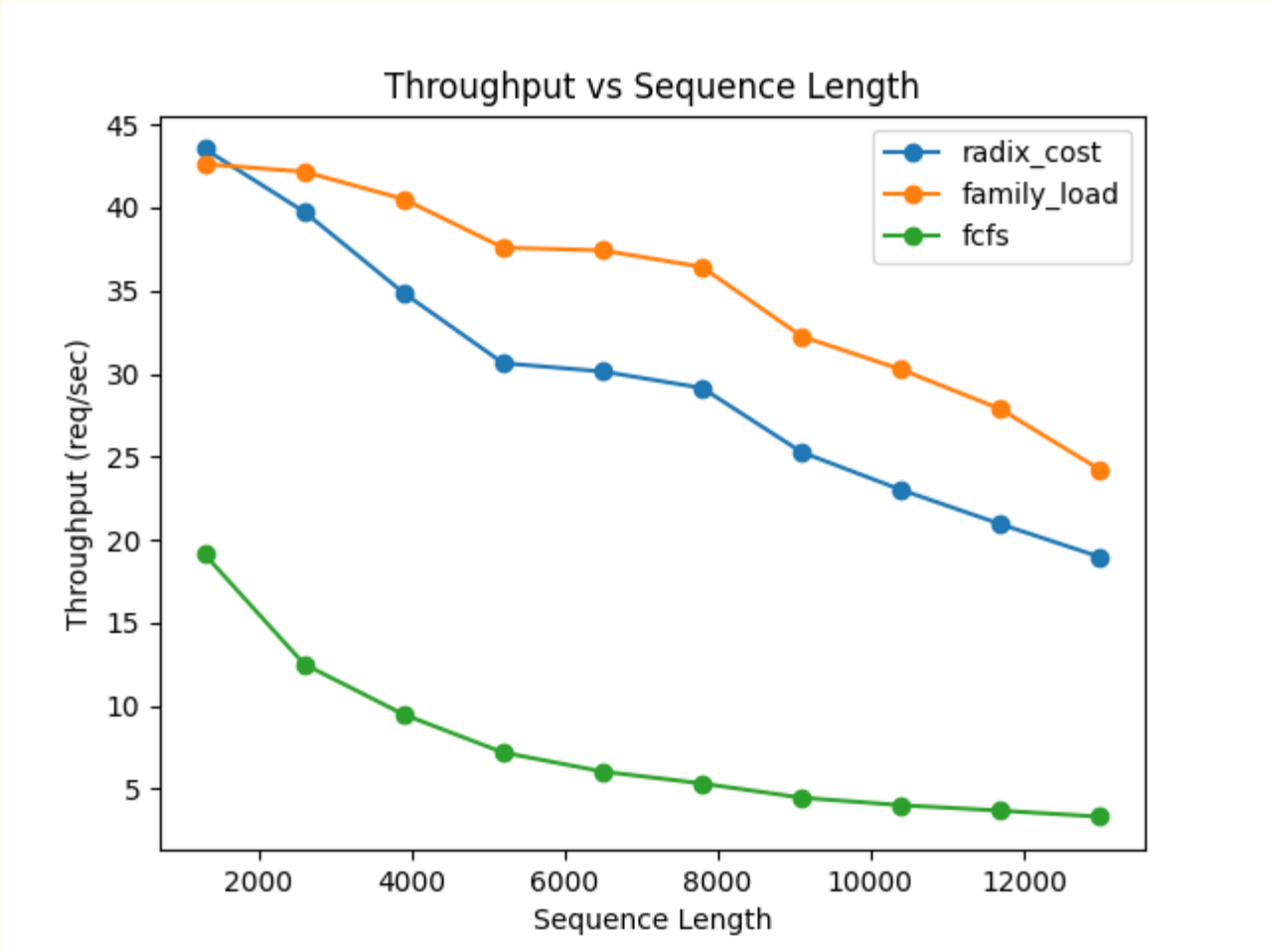


Figure: Throughput vs Sequence Length



Thank You